TFVP-MLLM: Training-Free Visual Prompting Multimodal Large Language Model for Remote Sensing Image Classification

Yifei Xu, Zixiang Ni, Pingping Wei, Chufan Xiao, Yaochen Li, Agen Qiu, Dong Li, and Yuan Rao

***Abstract*—The multimodal large language models (MLLMs) have achieved breakthroughs in remote sensing applications because of their powerful cross-modal understanding, enabling effective alignment between linguistic descriptions and visual data. However, the existing MLLM-based approaches still suffer from challenges, such as inaccurate region localization, limited domain adaptability, and high computational costs. To overcome these limitations, we propose a training-free visual prompting MLLM called TFVP-MLLM, which is built on visual prompting guidance and a scoring feedback mechanism to achieve high-quality remote sensing image classification under low resource constraints. Our TFVP-MLLM is composed of the category proposal generator (CPG) and the score feedback refiner (SFR). The CPG is responsible for generating high-quality candidate categories by integrating the proposed geographic priors injection module (GPIM) and hybrid-granularity region attention loss (HGRAL). Meanwhile, SFR uses an iterative generate and score process to enhance the accuracy and reliability of the candidate categories. As two critical components of CPG, GPIM incorporates prior geographic knowledge to significantly improve the model's domain adaptation, while HGRAL optimizes visual token representations with several frozen modules to reduce computational requirements. To validate the effectiveness of the proposed TFVP-MLLM, the comprehensive experiments are conducted on three benchmark datasets UCM, AID, and DIOR-RSVG. For the scene classification task, TFVP-MLLM achieves the state-of-the-art performance among training-free multimodal approaches with overall classification accuracies of 97.8% and 82.4% on the UCM and AID datasets, respectively. For the regional object classification task, TFVP-MLLM achieves remarkable scores of 92.4% semantic similarity (SS) and 88.6% semantic intersection over union (SIOU) on the DIOR-RSVG dataset, which surpasses the other training-free methods and even achieves comparable results against supervised approaches. In addition, we perform a series of ablation studies to confirm the effectiveness of each module. The source code is available at https://github.com/Camellia86/TFVP-MLLM**

***Index Terms*—Multimodal large language model (MLLM), remote sensing image classification, training-free, visual prompt.**

NOMENCLATURE

I	Input remote sensing image.
K	Global land cover categories percentage matrix $\mathbf{K} \in \mathbb{R}^{n \times m}$.
T	Original text prompt.
$\mathbf{T}_{\text{prompt}}$	Enhanced prompt formed by integrating **K** with T.
$\mathbf{P}_{\text{img}}$	Visual features $\mathbf{P}_{\text{img}} \in \mathbb{R}^d$ from frozen vision encoder.
$\mathbf{P}_{\text{text}}$	Textual features from token embedding of $\mathbf{T}_{\text{prompt}}$.
$\mathbf{P}_{\text{input}}$	Joint input representation combining $\mathbf{P}_{\text{img}}$ and $\mathbf{P}_{\text{text}}$.
A	Attention weight matrix $\mathbf{A} \in \mathbb{R}^{h \times s \times s}$ in frozen MLLM.
$\mathbf{I}_{\text{prompt}}$	Visual prompt used to guide attention weights.
$\mathcal{R}_{\text{cls}}$	Candidate classification set $\mathcal{R}_{\text{cls}} = \{R^1, R^2, \ldots, R^k\}$.
GI	Geospatial information vector $\mathbf{GI} = (GI^1, GI^2, GI^3, GI^4)$.
M	Land cover map from GlobeLand30 dataset.
l_i	Pixel-level indicator function for category i.
$Q(i)$	Quantity of category i in the land cover map **M**.
$B(i)$	Proportion of category i in the land cover map **M**.
$\mathbf{P}_{\text{v}}$	Learnable latent visual variable for attention optimization.
$\mathcal{L}_{\text{image}}$	Image-level loss.
$\mathcal{L}_{\text{obj}}$	Object-level loss.

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Yifei Xu, Zixiang Ni, Yaochen Li, and Yuan Rao are with the School of Software, Shaanxi Joint Key Laboratory for Artifact Intelligence, and Xi'an Key Laboratory for Social Intelligence and Complex Data Processing, Xi'an Jiaotong University, Xi'an, Shaanxi 710054, China, and also with Xi'an Jiaotong University Suzhou Institute, Suzhou, Jiangsu 215000, China (e-mail: belonxu_1@xjtu.edu.cn; nzx@stu.xjtu.edu.cn; yaochenli@xjtu.edu.cn; raoyuan@xjtu.edu.cn).

Pingping Wei is with the State Key Laboratory for Manufacturing Systems Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi 710054, China (e-mail: erin1989@xjtu.edu.cn).

Chufan Xiao is with XJTU-POLIMI Joint School, Xi'an Jiaotong University, Xi'an, Shaanxi 710054, China (e-mail: 3440653436@stu.xjtu.edu.cn).

Agen Qiu is with Chinese Academy of Surveying and Mapping, Beijing 100036, China (e-mail: qiuag@casm.ac.cn).

Dong Li is with the Department of Publicity, Xi'an Jiaotong University, Xi'an, Shaanxi 710054, China (e-mail: lidong05.03@xjtu.edu.cn).

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$\mathcal{L}_{\text{pixel}}$	Pixel-level loss.
$\mathcal{L}_{\text{total}}$	Total loss.
C	Predefined category list $C = \{C^1, C^2, \ldots, C^M\}$ in SFR.
R^i	A candidate category in SFR.
S^i	Similarity score S^i for candidate R^i from CLIP model.
$\mathbf{I}_c$	Cropped image region for candidate scoring.
Pred	Final classification result which represents the category of $\mathbf{I}$.

I. Introduction

REMOTE sensing image classification provides critical data for monitoring Earth's systems and managing resources [1], [2], [3]. In agricultural applications, classification algorithms support precision farming through crop type identification and health assessment, thereby supporting optimized irrigation and fertilization strategies to enhance yields [4], [5], [6]. Moreover, this technology is also vital for disaster management by facilitating rapid damage assessment after events, such as floods or wildfires and enabling more timely and effective emergency responses [7], [8], [9].

Deep learning has become a dominant approach in remote sensing image classification. As two classical representatives, the convolutional neural networks (CNNs) use localized filters to extract hierarchical spatial features, such as textures and patterns [10], [11], [12], and vision transformers (ViTs) utilize self-attention mechanisms to model long-range dependencies and capture global context across entire images [13], [14], [15], [16], [17]. Nevertheless, a critical limitation of both CNNs and ViTs lies in their primary focus on single-modality visual data, which fundamentally hinders their ability to resolve ambiguities in complex scenes where different targets exhibit similar visual characteristics [18].

Building upon the achievements of multimodal large language models (MLLMs) in traditional vision tasks [19], it has emerged as a very promising research direction to apply MLLM to remote sensing [20], [21], [22], [23]. For instance, VisualChatGPT [24] first explored LLM applications for remote sensing visual-language tasks, while Geochat [25] developed a multitask architecture for optical remote sensing images, and EarthGPT [26] constructed a generalized framework supporting a wide range of visual tasks. However, the existing MLLMs are confronted with three main challenges when processing remote sensing data.

1) *Inaccurate Region Localization:* The dense and unstructured spatial distribution of targets in remote sensing imagery results in overlapping and ambiguous boundaries. This complexity undermines the effectiveness of conventional text prompts for precise localization and consequently leads to diminished classification accuracy.
2) *Limited Domain Adaptation:* Most MLLMs are pretrained on natural images rather than remote sensing data, which leads to their insufficient capacity for domain adaptation.
3) *High Computational Costs:* The extensive computational burden of processing large-scale remote sensing datasets poses a significant bottleneck, thus impeding model optimization and practical real-world deployment.

To solve the above problems, we propose a training-free MLLM named TFVP-MLLM to leverage visual prompting guidance for remote sensing image classification. The TFVP-MLLM framework consists of the category proposal generator (CPG) and the score feedback refiner (SFR). The CPG integrates the geographic priors injection module (GPIM) and the hybrid-granularity region attention loss (HGRAL) to generate high-quality candidate categories. Specifically, the GPIM enhances the model's domain adaptation by leveraging the GlobeLand30 dataset to integrate real-world geographic knowledge into text prompts. This provides crucial prior knowledge that significantly improves the model's adaptability to diverse remote sensing scenarios. Meanwhile, the HGRAL avoids fine-tuning the model layer by directly adjusting the visual tokens. This strategy optimizes the attention map through hybrid-granularity loss, enabling precise focus on key image regions and achieving robust visual-language alignment at substantially lower computational costs. Through these two modules, the CPG iteratively optimizes attention distribution for improving candidate category generation. Moreover, the SFR introduces a scoring-feedback mechanism to remote sensing, which first filters the candidate categories proposed by the CPG, and then cyclically evaluates and refines them through iterative feedback. The experimental results demonstrate that TFVP-MLLM performs effectively in scene classification and regional object classification. Furthermore, the ablation studies are conducted to the essential components of our proposed model. In summary, the main contributions of our study are as follows.

1) A pioneering training-free MLLM framework termed TFVP-MLLM is proposed for remote sensing image classification. This approach enables rapid adaptation to remote sensing domain-specific tasks without fine-tuning the model while maintaining high classification accuracy.
2) The CPG first leverages the GPIM to inject real-world geographic knowledge for enhanced domain adaptation and then employs the HGRAL to achieve precise visual-language alignment at minimal computational cost via hybrid-granularity attention optimization. Finally, the SFR introduces a scoring-feedback mechanism to filter and iteratively refine the generated candidate categories.
3) Extensive experimental evaluations demonstrate the effectiveness of the proposed framework. TFVP-MLLM achieves competitive performance across multiple benchmark datasets, including scene classification tasks on UCM and AID datasets as well as regional object classification task on DIOR-RSVG.

The remainder of this article is organized as follows. A brief review of the related work is provided in Section II. Section III details the proposed method. Experimental results on benchmark datasets along with ablation studies are presented in Section IV. Finally, Section V concludes this article. The descriptions of notations used throughout this article are given in Nomenclature.

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II. Related Work

In this section, we will review the research trends of MLLM in remote sensing image classification task from three different perspectives: MLLMs, training-free in MLLMs, and visual prompts in MLLMs.

A. MLLM

The MLLMs typically consist of a visual encoder, an LLM decoder, and an image–text alignment module. MLLMs have demonstrated exceptional applicability in remote sensing due to their robust cross-modal comprehension capabilities, which facilitate the seamless integration of visual and linguistic data processing [21], [27], [28], [29], [30], [31], [32], [33]. The pioneering work of VisualChatGPT [24] first revealed the potential to use LLMs for visual-language tasks in remote sensing. Building upon this, GeoChat [25] emerged as the first MLLM specifically designed to address multiple tasks on remote sensing imagery. Subsequently, EarthGPT [26] proposed a unified MLLM framework capable of handling multimodal remote sensing data across a wide spectrum of visual tasks. To tackle the challenges related to visual information compression and cross-modal alignment, LHRS-Bot-Novacitebib:34 introduced enhanced visual encoding and connector mechanisms in an efficient way. As the first vision-language foundation model, RemoteCLIP [21] aims to learn robust visual features with rich semantics, as well as aligned text embeddings for seamless downstream application. Recently, EarthMarkercitebib:35 pioneered the use of visual prompting guidance for remote sensing image understanding within an LLM framework. Although these advances have markedly advanced the application of MLLMs in remote sensing, the existing methods typically require training from scratch or involve extensive fine-tuning. Consequently, such approaches often entail high computational demands and considerable training expenses.

B. Training-Free in MLLMs

Training-free approaches in MLLMs have primarily progressed along two primary directions. One line of work focuses on lightweight internal interventions to harness the model's inherent knowledge for controllable generation [36], [37], [38], [39], [40]. As one of the representative approaches, ControlMLLMcitebib:41 steered the model's attention through visual prompts to adjust latent visual variables and activate inherent capabilities. Concurrently, another direction couples MLLMs with external models in a training-free manner to provide enhanced perceptual capabilities, specialized knowledge, and iterative refinement mechanisms [42], [43], [44], [45], [46]. For instance, MILS [47] embedded an MLLM within a generate-evaluate loop alongside pretrained models to iteratively refine outputs. Collectively, these efforts highlight the growing relevance and promise of training-free strategies in enhancing both the flexibility and practicality of MLLM applications.

Algorithm 1 Procedure of TFVP-MLLM

Input: The remote sensing image $\mathbf{I}$, The initial text prompt T
Output: The final classification result **Pred**

Step 1. Category Proposal Generator:
  Retrieve $\mathbf{K}$ from $\mathbf{I}$ using GPIM
  Compute $\mathbf{T}_{\text{prompt}}$ from $\mathbf{K}$ and T
  Compute $\mathbf{P}_{\text{text}}$ from $\mathbf{T}_{\text{prompt}}$ using token embedding
  Compute $\mathbf{P}_{\text{img}}$ from $\mathbf{I}$ using frozen MLP and visual encoder
for J_1 iterations **do**
    Form $\mathbf{P}_{\text{input}}$ using $\mathbf{P}_{\text{img}}$ and $\mathbf{P}_{\text{text}}$
    Extract $\mathbf{A}$ from $\mathbf{P}_{\text{input}}$ using LLM
    Compute $\mathbf{P}_{\text{v}}$ from HGRAL using Eq. 10 to Eq. 14
    Update $\mathbf{P}_{\text{img}}$ using $\mathbf{A}$ and $\mathbf{P}_{\text{v}}$
end for
  Obtain the Candidate Set $\mathcal{R}_{\text{cls}}$
Step 2. Scoring Feedback Refiner:
  (1) Filter $\mathcal{R}_{\text{cls}}$ using Eq. 15
  (2) Score candidate categories:
for each $R_i \in \mathcal{R}_{\text{cls}}$ **do**
    Compute S^i using Eq. 16
end for
  (3) Update text prompt T
3. Iterative Refinement
for J_2 iterations **do**
    Repeat **Steps 1-2**
end for
Compute the final Classification result **Pred** using Eq. 17

C. Visual Prompting in MLLMs

As a crucial bridge connecting the visual modality with semantic understanding, visual prompting techniques have demonstrated significant guidance effects on various multimodal tasks [48], [49], [50], [51], [52], [53], [54], [55], [56]. ConMLLM [41] and Sphinx [57] enhanced MLLMs' region recognition by jointly encoding spatial frames and ROI features with textual inputs. Osprey [58] incorporated masked regions into linguistic command to enable pixel-level understanding, while EarthVQA [59] integrated images, visual prompts, and QA pairs for object-centric reasoning. For remote sensing imagery, RSPrompter [52] introduced an automated prompt generation method to adapt segment anything model (SAM) [60] for segmentation tasks. However, these methods are often limited to specific applications and require task-specific fine-tuning with downstream data.

III. Methodology

The overall network architecture of TFVP-MLLM is shown in Fig. 1. It mainly consists of CPG and SFR. The CPG processes the remote sensing image $\mathbf{I}$ through GPIM to retrieve a global land cover categories percentage matrix $\mathbf{K} \in \mathbb{R}^{n\times m}$, which is combined with the original text prompt T to form an enhanced prompt $\mathbf{T}_{\text{prompt}}$. Simultaneously, the input image $\mathbf{I}$ is encoded via a frozen visual encoder followed by a frozen MLP to produce visual tokens $\mathbf{P}_{\text{img}} \in \mathbb{R}^d$, while $\mathbf{T}_{\text{prompt}}$ is tokenized into text prompt tokens $\mathbf{P}_{\text{text}}$. Afterward, $\mathbf{P}_{\text{text}}$ and $\mathbf{P}_{\text{img}}$ are concatenated into a joint input representation $\mathbf{P}_{\text{input}}$ as the input of subsequent frozen LLM. Rather than directly outputting candidate classification results, the HGRAL is proposed to enhance inference through visual prompting by extracting and modulating the attention map $\mathbf{A} \in \mathbb{R}^{h\times w\times s}$ from the frozen LLM. Following J_1 inner loops, a candidate classification

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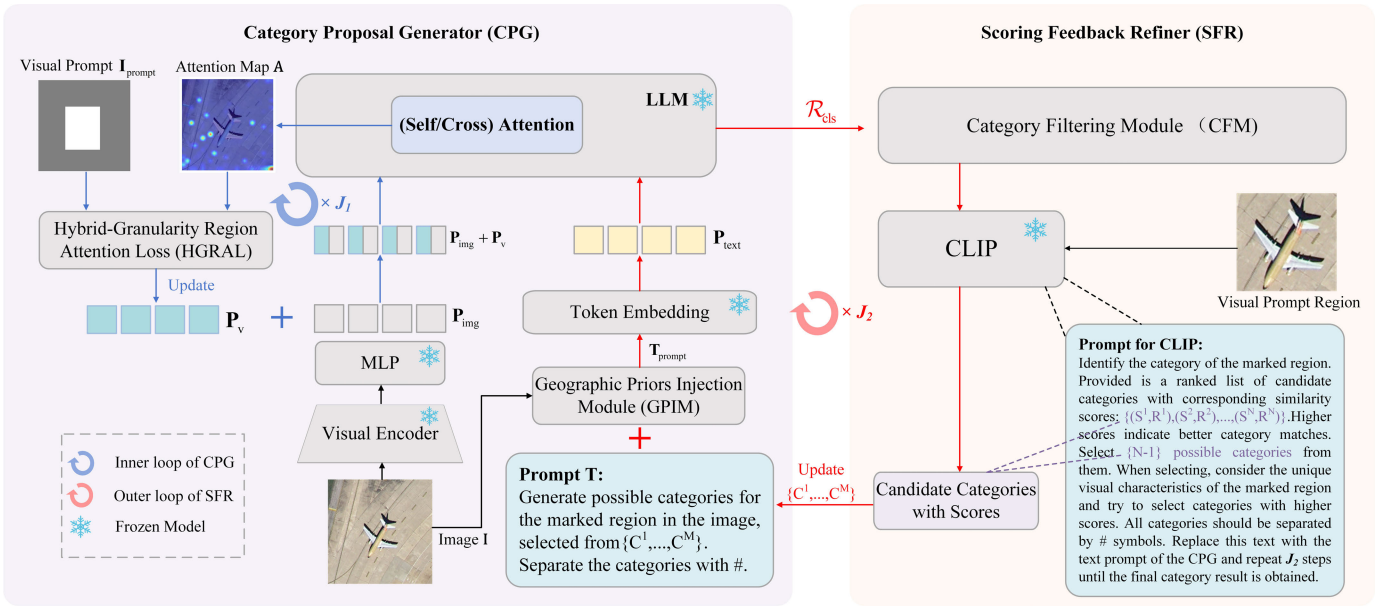


Fig. 1. Overview of the proposed TFVP-MLLM framework. The text prompt T is combined with the global land cover categories percentage matrix **K** retrieved by GPIM to form enhanced prompt $\mathbf{T}_{prompt}$, which is then tokenized into text prompt tokens $\mathbf{P}_{text}$. Meanwhile, the input image **I** is processed by frozen visual encoder and MLP to produce visual tokens $\mathbf{P}_{img}$. Both $\mathbf{P}_{img}$ and $\mathbf{P}_{text}$ are inputted to the frozen LLM. Guided by the visual prompt $\mathbf{I}_{prompt}$, the HGRAL iteratively adjusts the attention map **A** from LLM to direct more focus to the prompted mask region. After J_1 inner loops (blue arrows), the LLM produces a candidate set $\mathcal{R}_{cls}$, which is first refined by the CFM and then scored by a CLIP-based model. Following J_2 outer loops (red arrows), the category with the highest confidence score is selected as the final prediction **Pred**.

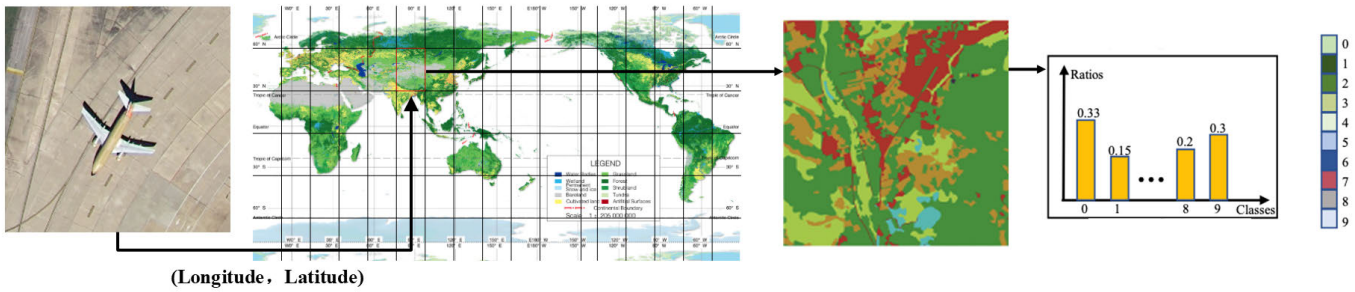


Fig. 2. Pipeline of GPIM. The GPIM dynamically queries land cover proportions using the input image's geographic coordinates and integrates this geographic knowledge by converting the statistics into terrain-aware prompt embedding.

set $\mathcal{R}_{cls} = \{R^1, R^2, \ldots, R^k\}$ is produced. The candidate set $\mathcal{R}_{cls}$ is then evaluated by SFR, where the category filtering module (CFM) first filters out incompatible candidates to reduce its size, and CLIP subsequently computes confidence scores to retain semantically consistent and high-confidence categories. These refined categories replace the preselected ones $C = \{C^1, C^2, \ldots, C^M\}$ in the original text prompt T of the CPG. After J_2 outer loops, SFR outputs the category with the highest score as the final prediction **Pred** through visual prompting guidance and scoring feedback. The procedure of the proposed TFVP-MLLM can be found in Algorithm 1.

A. GPIM

As illustrated in Fig. 2, the goal of GPIM is to integrate geographic knowledge into remote sensing image analysis to improve visual reasoning with structured prior information. A key motivation is that the regions with similar land cover compositions tend to exhibit consistent feature representations, making land cover statistics a reliable form of domain prior [61]. To operationalize this idea, we leverage the GlobeLand30 dataset as it provides comprehensive coverage of ten standardized land cover categories at a 30-m resolution across global geographic regions. In detail, GPIM first queries the proportional distribution of land cover categories from GlobeLand30 corresponding to the geographic location of the input image **I** and then transforms these statistics into terrain-aware prompt embedding. The above process allows GPIM to effectively bridge the gap between low-level image features and high-level geographic semantic attributes.

Formally, the geospatial information of input **I** is represented as $\mathbf{GI} = (GI^1, GI^2, GI^3, GI^4)$, where (GI^1, GI^2) are the longitude and latitude coordinates of the top-left corner, and (GI^3, GI^4) represent its horizontal and vertical spatial resolutions, respectively. In the case of remote sensing images with native geographic registration, we utilize GDAL library functions to directly extract strictly aligned geographic coordinates (including the four corners and center latitude/longitude

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of the coverage area) and spatial resolution from the image header files. Conversely, for public datasets that lack such information, we augment them with the necessary geographic metadata by leveraging the global semantic-rich LHRS-Align dataset [34] and the geo-alignment pipeline established in this study. After obtaining this geospatial data, the target region GI_{tg} is located within the GlobeLand30 dataset. Subsequently, the relative position of the remote sensing image inside GI_{tg} is derived from the following equations:

$$GI_{tg}^{\text{left}} = \left\{GI^1 - GI_{tg}^1\right\} / GI_{tg}^3 \tag{1}$$

$$GI_{tg}^{\text{top}} = \left\{GI^2 - GI_{tg}^2\right\} / GI_{tg}^4 \tag{2}$$

$$GI_{tg}^{\text{right}} = GI_{tg}^{\text{left}} + \left(GI^3 \times w\right) / GI_{tg}^3 \tag{3}$$

$$GI_{tg}^{\text{bottom}} = GI_{tg}^{\text{top}} + \left(GI^4 \times h\right) / GI_{tg}^4 \tag{4}$$

where $(GI_{tg}^{\text{left}}, GI_{tg}^{\text{top}})$ and $(GI_{tg}^{\text{right}}, GI_{tg}^{\text{bottom}})$ denote the top-left and bottom-right corner coordinates of the image within target region GI_{tg}. h and w are the image height and width, respectively. Using the positional coordinate information $(GI_{tg}^{\text{left}}, GI_{tg}^{\text{top}})$ and $(GI_{tg}^{\text{right}}, GI_{tg}^{\text{bottom}})$, both the quantity $Q(\cdot)$ and proportion $B(\cdot)$ of each category within the target region are computed as

$$Q(i) = \sum_{x,y} l_i\left(\mathbf{M}(x,y)\right) \tag{5}$$

$$B(i) = \frac{Q(i)}{\sum_j Q(j)} \tag{6}$$

$$l_i = 1 \quad if\ \mathbf{M}(x,y) = i \quad \text{else } 0 \tag{7}$$

where i denotes a land cover category with $l_i(\cdot)$ being its pixel-level indicator function, $\mathbf{M}$ represents the land cover map from GlobeLand30 dataset, and $\mathbf{M}(x,y)$ is the land cover category at the location of the coordinates (x,y). $B(\cdot)$ calculates the proportion area of each category within the target region derived from the land cover map of GI_{tg} to obtain categories percentage matrix $\mathbf{K}$. Afterward, $\mathbf{K}$ and T are combined to form the enhanced prompt $\mathbf{T}_{\text{prompt}}$, which is then fed into the frozen token embedding module to output $\mathbf{P}_{\text{text}}$.

B. HGRAL

Inspired by ControlMLLM [41], our proposed HGRAL optimizes visual features using latent visual variables. To further enhance the perception of critical regions, HGRAL designs a hybrid-granularity loss to directly refine these variables in the attention map to achieve precise region guidance without training.

Specifically, the mask region is first transformed to binary mask $\mathbf{I}_{\text{prompt}}$. Currently, the learnable latent visual variable $\mathbf{P}_{\text{v}}$ is initialized with the same size as the visual token $\mathbf{P}_{\text{img}}$ and add it to $\mathbf{P}_{\text{img}}$. Those visual tokens along with the text prompt token $\mathbf{P}_{\text{text}}$ are fed into the LLM to optimize the attention distribution within its attention layers. The attention map $\mathbf{A}^{(\tau)}$ generated at layer τ is computed as follows [62]:

$$\mathbf{A}^{(\tau)} = \text{softmax}\left(\frac{[e_v, e_t]^{(\tau)} \cdot \left([e_v, e_t]^{(\tau)}\right)^T}{\sqrt{d_k}} + Ct\right) \tag{8}$$

$$Ct_i = \eta \quad \text{if } i \in N_r \quad \text{else } 0 \tag{9}$$

where d_k is a scaling factor, N_r denotes the number of pixels inside $\mathbf{I}_{\text{prompt}}$, e_v and e_t represent the visual token $\mathbf{P}_{\text{img}} + \mathbf{P}_{\text{v}}$ and text token $\mathbf{P}_{\text{text}}$ inputted into the LLM, respectively, and they are computed on the transformed visual-text concatenated embeddings $[e_v, e_t]$. η is an edit attention factor [63], which is set to 10. $\mathbf{A}_{ij}^{(\tau)}$ represents the relationship between the ith token and the jth token. Then, the attention map $\mathbf{A}$ is calculated by averaging pool the attention maps generated for each text prompt token to represent the global context of the text prompt and its association with visual tokens. Building upon these representations, HGRAL is designed to update $\mathbf{P}_{\text{v}}$, which is composed of image-level loss $\mathcal{L}_{\text{image}}$, object-level loss $\mathcal{L}_{obj}$, and pixel-level loss $\mathcal{L}_{\text{pixel}}$.

The image-level loss sharpens the global attention contrast between the masked foreground and background to improve the saliency of targets and suppress background interference. Its definition can be expressed as

$$\mathcal{L}_{\text{image}} = \begin{cases} -\log\left(\dfrac{\frac{1}{|N_r|}\sum_{i\in N_r}\mathbf{A}_i^{\text{mask}} + \epsilon}{\frac{1}{|N_{\bar{r}}|+\epsilon}\sum_{i\notin N_r}\mathbf{A}_i^{\text{mask}} + \epsilon}\right), & \text{when} |N_{\bar{r}}| \neq 0 \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where N_r and $N_{\bar{r}}$ denote the number of pixels inside and outside the $\mathbf{I}_{\text{prompt}}$, respectively. $\mathbf{A}^{\text{mask}}$ is the attention map within the masked region. To ensure numerical stability, $\epsilon = 1 \times 10^{-6}$ acts as a smoothing term. When $N_{\bar{r}} = 0$, the image loss term $\mathcal{L}_{\text{image}}$ is discarded.

The object-level loss $\mathcal{L}_{obj}$ employs a local variance minimization strategy [see (11)] to distribute attention evenly within the mask region. By minimizing the variance of attention values, it prevents overfocusing on local patches and ensures a consistent response across the entire mask

$$\mathcal{L}_{\text{obj}} = \frac{1}{|N_r|}\sum_{i\in N_r}\left(\mathbf{A}_i - \frac{1}{|N_r|}\sum_{i\in N_r}\mathbf{A}_i\right)^2. \tag{11}$$

The pixel-level loss employs a Gaussian distance weighting scheme to enforce stronger attention focus within mask region, where attention intensity increases with proximity to the mask boundaries

$$\mathcal{L}_{\text{pixel}} = \left(1 - \frac{\sum_{i\in N_r}\frac{e^{-D_i^2/2\sigma^2}}{\sqrt{2\pi}\sigma}\mathbf{A}_i}{\sum_i \mathbf{A}_i}\right)^2 \tag{12}$$

where D_i measures the Euclidean distance from pixel i to the mask region, and $\sigma = 0.1$ is the Gaussian standard deviation. It is worth noting that when the mask covers the entire image (as in scene classification, where $N_{\bar{r}} = 0$), $\mathcal{L}_{\text{image}}$ becomes zero according to (10) and the distance-based Gaussian weighting in $\mathcal{L}_{\text{pixel}}$ reduces to a spatial constant. However, $\mathcal{L}_{\text{obj}}$ remains fully operational: it minimizes the variance of attention weights across all spatial positions, encouraging the attention map to distribute uniformly over the entire scene rather than concentrating on local textures. The total loss combines three-level losses as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{image}} + \mathcal{L}_{\text{obj}} + \mathcal{L}_{\text{pixel}}. \tag{13}$$

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The gradient update rule for learnable latent visual variable $\mathbf{P}_v$ is

$$\mathbf{P}_v \leftarrow \mathbf{P}_v - \alpha \nabla_{\mathbf{P}_v} \mathcal{L}_{\text{total}}(\mathbf{A}, N_r) \tag{14}$$

where α is the hyperparameter controlling the guidance strength. By optimizing $\mathbf{P}_v$, it indirectly guides the attention map to produce stronger responses in the reference region r, thereby enhancing the influence of visual prompts on the output. Finally, the learnable latent visual variable $\mathbf{P}_v$ undergoes J_1 iterative optimization steps. To enhance model stability and prevent overfitting, the early stopping strategy is utilized.

C. Scoring Feedback Refiner

The SFR implements an iterative generate-score feedback loop to refine the candidate categories $\mathcal{R}_{\text{cls}}$ output by the CPG. The SFR comprises the CFM and a CLIP-based scorer. The CFM ensures output compliance by first verifying candidate answers against predefined classes and then mapping out-of-distribution candidates to their closest semantic matches within valid categories while discarding irreconcilable outliers, thereby guaranteeing both standardization and practical utility of final outputs.

The SFR implements an iterative generate-score feedback loop to refine the candidate categories $\mathcal{R}_{\text{cls}}$ output by the CPG. It consists of a CLIP-based visual scorer and a CFM. The CLIP-based scorer calculates the visual-semantic similarity (SS) between each candidate category and the input image (or the cropped target region for regional tasks) and then feeds the similarity scores back to the frozen LLM in CPG to guide it to generate more accurate candidate sets. This process repeats for a predefined number of iterations to progressively filter out low-confidence predictions and optimize the ranking of valid candidates. Finally, the CFM verifies that all remaining candidates fall within the predefined task class set, maps out-of-distribution candidates to their closest semantic matches, and discards irreconcilable outliers to ensure standardized outputs.

As illustrated in Fig. 1, the initial text prompt for the CPG is "Select N possible categories from the {category} list for the image region and separate them with # symbols." The CPG produces N candidate categories based on this prompt and outputs the candidate category set $\mathcal{R}_{\text{cls}} = \{R^1, R^2, \ldots, R^N\}$. Then, the candidate set is processed by the CFM. Let the predefined category list be $C = \{C^1, C^2, \ldots, C^M\}$. According to (15), a verification is first performed to check if the candidate result R^i is a member of C. If not, the SS between R^i and all categories in C is computed using Sentence-BERT [64], and the maximum similarity score $\text{sim}_{\max}$ is identified. If $\text{sim}_{\max}$ is greater than or equal to θ_{sim}, R^i is corrected to the corresponding best-matching category C^{best}. Otherwise, the candidate result is discarded

$$R^i_{\text{final}} = \begin{cases} R^i, & \text{if } R^i \in C \\ C^{\text{best}}, & \text{if } R^i \notin C \text{ and } \text{sim}_{\max} \geq \theta_{sim} \\ \text{discarded}, & \text{otherwise.} \end{cases} \tag{15}$$

The filtered categories are then scored by a CLIP [65] to evaluate the semantic affinity between each candidate category R^i and the cropped marked region image from $\mathbf{I}_c$ to yield a score S_i

$$S^i = \text{CLIP}\left(R^i, \mathbf{I}_c\right). \tag{16}$$

The prompt for CLIP to score candidate categories is as follows: "Identify the category of the marked region. Provided is a ranked list of candidate categories with corresponding similarity scores: {(S^1,R^1), (S^2,R^2), …, (S^N,R^N)}. Higher scores indicate better category matches. Select N-1 possible categories from them. When selecting, consider the unique visual characteristics of the marked region and try to select categories with higher scores. All categories should be separated by # symbols. Replace this text with the text prompt of the CPG and repeat J_2 steps until the final category result is obtained." Hence, the final classification result **Pred** is computed as follows:

$$\textbf{Pred} = \arg\max_{S^i} R^i. \tag{17}$$

TABLE I
SUMMARY OF EXPERIMENTAL DATASETS. ABBREVIATION: CLASSIFICATION (CLS)

Dataset	Task Type	Categories	Images	Image Size
UCM	Scene Cls	21	2100	256×256
AID	Scene Cls	30	10000	Multi-scale
DIOR-RSVG	Region Cls	20	3375	800×800

IV. EXPERIMENTS

A. Datasets

To systematically evaluate the performance of the proposed TFVP-MLLM, we conducted comprehensive experiments on three representative multimodal remote sensing datasets: UCM [66], AID [67], and DIOR-RSVG [68]. The UCM and AID datasets were utilized to assess scene classification performance in scenarios lacking geographic metadata, while the DIOR-RSVG dataset was employed to evaluate regional object classification capabilities leveraging its native geographic registration information. The details of these datasets are summarized in Table I.

1) UCM Dataset: The UCM dataset was developed by the University of California, Merced, using imagery from the USGS National Map Urban Area Imagery Collection. It contains 2100 standardized remote sensing images. Each image has a size of 256 × 256 pixels and a spatial resolution of 0.3 m per pixel. The dataset includes 21 scene classes covering diverse land cover categories, such as residential areas, transportation facilities, and natural landscapes.

2) AID Dataset: The AID dataset was collected from the Google Earth platform and released by Wuhan University. It consists of 10 000 multiresolution remote sensing images ranging from 0.5 to 8 m per pixel across 30 semantic categories. The dataset features global coverage and incorporates natural variations in season, lighting, weather, and observation perspective.

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3) DIOR-RSVG Dataset: As an extended version of DIOR dataset, the vision-language multimodal DIOR-RSVG dataset was developed by Northwestern Polytechnical University. It contains 17 402 remote sensing images of 800 × 800 pixels ranging from 0.5 to 30 m per pixel across 20 categories. Besides, it provides image–text correspondences containing metadata, bounding boxes, and labels.

B. Experimental Details

1) Experimental Environment: All the models were trained on an Ubuntu 20.04 operating system with two NVIDIA RTX 3090 GPUs with 24-GB memory. All experiments are implemented by PyTorch. All experiments, including the training of supervised baselines and the inference of training-free models, were conducted on an Ubuntu 20.04 operating system equipped with two NVIDIA RTX 3090 GPUs (24-GB memory). All algorithms were implemented using PyTorch.

2) Implementation Details: In this study, LLaMA2-7B-Chat serves as the frozen LLM in CPG, while clip-vit-large-patch14 functions as the CLIP scorer in SFR. The remaining frozen modules, including the visual encoder, MLP, and token embedding with frozen weights, are sourced from LHRS-Bot [34]. In the experiments, the number of candidate categories N is set to 3 and SFR is repeated $J_2 = 2$ times. For the regional object classification task, HGRAL is repeated $J_1 = 3$ times, the guidance strength control parameter α is 500. θ_{sim} in (15) is set to 0.5. Following the definition in Section III-B, the mask region covers the entire image extent [0, 0, w, h] for scene classification, whereas it corresponds to the preannotated bounding boxes from the DIOR-RSVG dataset for regional object classification. All main experiments and comparisons with the existing methods use CLIP-ViT-L/14 as the default scorer to ensure fair comparison. For the ablation study on scorer domain specificity, RemoteCLIP-ViT-L/14 [21] is evaluated as a replacement scorer within SFR while keeping all other components identical (see Section IV).

3) Evaluation Metrics: For scene classification task, we adopt overall accuracy (OA), average accuracy (AA), and Kappa Coefficient (κ) to make quantitative analysis of different methods comprehensively and fairly. Notably, we report the results over accuracy for each category in UCM and AID datasets.

The goal of regional object classification is to identify the categories within a specified region of an image. The evaluation metrics include SS and semantic intersection over union (SIoU) [69]. SS measures the similarity between the predicted label and its ground truth in embedding space. SIoU evaluates the regional overlap between the predicted and ground-truth labels.

4) Comparative Approaches: To evaluate the performance of TFVP-MLLM, we select the following baseline methods for comparative experiments.

1) Qwen-VL-Chat [70] is a general-purpose 7B parameters MLLM that excels at vision-language alignment, multi-image and multiturn dialog, visual question answering, OCR, and fine-grained visual tasks.
2) MiniGPTV2 [71] is a zero-shot lightweight multimodal reasoning framework based on LLaMA2-7B-Chat that adopts a generator-scorer iterative mechanism to achieve description and cross-modal reasoning for images, videos, and audio.
3) LLaVA-1.5 [72] is a general-purpose 13B parameters MLLM that utilizes a CLIP-ViT-L-14 visual encoder and incorporates an MLP-based cross-modal connector. It supports high-quality image captioning, visual question answering, OCR, and cross-lingual dialog.
4) Sphinx [57] is an LLaMA2-7B-based MLLM with a mixed strategy of weights, visual embeddings, and image scales for high-resolution image task.
5) ControlMLLM [41] is a training-free MLLM based on LLaVA-1.5-7B that optimizes attention maps using latent visual variables. It is applicable to the majority of vision-language alignment tasks, particularly those under limited computational resources.
6) GeoChat [25] is a remote sensing-specialized MLLM built on Vicuna-v1.5-7B and further enhanced via instruction tuning specifically tailored for remote sensing data. It supports image-level, region-level, and embodied dialog functionalities.
7) LHRS-Bot-Nova [34] is a remote sensing-specialized MLLM based on Vicuna-v1.5-7B that employs a high-resolution visual encoder and a mixture-of-experts bridging layer for instruction tuning. It is suitable for tasks, such as image classification, visual question answering, and spatial localization in remote sensing scenarios.
8) RemoteCLIP [21] is a remote sensing-specialized visual encoder based on ViT-L/14 that leverages contrastive language-image pretraining on large-scale collections of remote sensing image–text pairs.
9) EarthMarker [35] is a remote sensing-specialized MLLM based on LLaMA2-13B that adopts a vision-language alignment method designed for remote sensing imagery. It enables interactive multiturn dialog, fine-grained description, and attribute extraction via visual prompting.

C. Comparisons

1) Quantitative Analysis: TFVP-MLLM and all baselines were evaluated on the UCM dataset, AID dataset, and DIOR-RSVG dataset for comparative analysis. For the scene classification task, the evaluation metrics include OA, AA, κ, and per-class classification accuracy.

Tables II and III report the overall classification accuracy and per-class accuracy on the UCM and AID datasets. In the scene classification, Sphinx demonstrates no significant accuracy improvement over Qwen-VL-Chat and LLaVA-1.5. This observation can be attributed to the nature of scene classification task, where the entire image serves as the region of interest to the extent of diminishing the impact of visual embeddings on model performance. Also, a pattern that is similarly observed in the remote sensing domain. Benefiting from richer domain knowledge, remote sensing-specialized MLLM exhibits superior zero-shot performance compared to general-purpose vision-language models. Among the training-free models with multiround iterative optimization, MILS and the

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TABLE II

COMPARISON OF SCENE CLASSIFICATION RESULTS FOR VARIOUS METHODS ON THE UCM DATASET (%). THE BEST PERFORMANCE IN EACH CATEGORY IS HIGHLIGHTED IN BOLD. ABBREVIATIONS: QWEN-VL-CHAT (QWEN), MINIGPT-V2 (MGPT), LLAVA-1.5 (LLAVA), CONTROLMLLM (CONMLLM), LHRS-BOT-NOVA (LHRS), EARTHMARKER (EMKER), REMOTECLIP (RCLIP), AND TFVP-MLLM (TMLLM). THE TYPES OF ALL COMPARATIVE MODELS ARE DENOTED AS: G: GENERAL IMAGE DOMAIN, RS: REMOTE SENSING DOMAIN, VP: VISUAL PROMPTING, AND TF: TRAINING-FREE

Category	G			G&VP	G&TF	G&VP&TF	RS			RS&VP	RS&VP&TF
	Qwen	MGPT	LLaVa	Sphinx	MILS	ConMLLM	GeoC	LHRS	RCLIP	EMker	TMLLM
OA	62.8	4.9	53.7	62.9	71.3	67.9	84.4	96.1	95.8	86.5	**97.8**
AA	62.8	4.9	53.7	62.9	71.3	67.9	84.4	96.1	95.7	86.5	**97.8**
κ	61	0.04	51.4	60.9	69.9	66.4	83.6	95.9	95.6	84.2	**97.7**

TABLE III

COMPARISON OF SCENE CLASSIFICATION RESULTS FOR VARIOUS METHODS ON THE AID DATASET (ACCURACY IN %)

Category	G			G&VP	G&TF	G&VP&TF	RS			RS&VP	RS&VP&TF
	Qwen	MGPT	LLaVa	Sphinx	MILS	ConMLLM	GeoC	LHRS	RCLIP	EMker	TMLLM
OA	52.6	12.9	51	58.2	65.8	55.3	72.1	81.5	**87.9**	77.9	82.4
AA	52.1	12.8	53.3	58	63.7	54.5	70.8	79.2	**87.2**	77.7	82.1
κ	50.8	9.9	49.5	56.5	63.8	52.5	70.5	80.6	**87.0**	77.1	81.8

proposed TFVP-MLLM achieve higher classification accuracy. Compared to LHRS-Bot-Nova [34], TFVP-MLLM improves accuracy by 1.7% (96.1%→97.8%) on UCM, which achieves the best performance in scene classification on UCM. On the AID dataset, our TFVP-MLLM attains 82.40% accuracy, which is lower than the 87.90% achieved by RemoteCLIP. In contrast, our approach demonstrates superior performance on the UCM dataset by reaching 97.8%, significantly outperforming the 95.8% result of RemoteCLIP significantly lower than our method's accuracy of 97.8%. This divergence stems from the distinct nature of the datasets and model architectures. RemoteCLIP benefits from extensive pretraining on 165 745 images with 828 725 corresponding image–text pairs, granting it robust priors for the complex multiresolution variations found in AID. However, this accuracy comes at the cost of 233.4 h of training on 4 NVIDIA 3090Ti GPUs. Conversely, our method achieves superior results on the UCM dataset by leveraging a strictly training-free paradigm. Instead of relying on massive domain-specific pretraining, TFVP-MLLM utilizes a frozen general-purpose LLaMA2-7B-Chat enhanced by our GPIM and SFR. These components dynamically refine visual-language alignment and incorporate real-world geographic knowledge, allowing our method to adapt effectively to the uniform high-resolution imagery of UCM. Consequently, TFVP-MLLM offers a highly efficient alternative that maintains high accuracy with significantly lower computational overhead and GPU memory usage compared to heavily pretrained foundation models.

As shown in Table IV, we evaluate the semantic comprehension capabilities of different MLLMs on the DIOR-RSVG dataset. Among the models specifically designed for remote sensing, EarthMarker achieves the best performance with SS and SIoU scores of 98.37% and 97.24%, respectively. Compared to Qwen-VL-Chat, Sphinx with visual prompts improves the SS metric by 19.2% (74.5%→93.7%). This indicates that models with stronger vision-language alignment capabilities hold a distinct advantage in understanding the semantics of remote sensing imagery. Simultaneously, the visual-prompted training-free ControlMLLM also obtains higher SS and SIoU scores than Qwen-VL-Chat. These stronger visual-language alignment capabilities (EarthMarker, Sphinx, and TFVP-MLLM) bring superior semantic understanding of remote sensing image. For our proposed TFVP-MLLM, it shows slightly lower SS and SIoU scores than EarthMarker and Sphinx in regional object classification. This performance difference is mainly attributed to the fact that EarthMarker and Sphinx are fully supervised models that have undergone extensive task-specific training on large-scale remote sensing visual prompting datasets, while our TFVP-MLLM operates in a completely training-free manner without any parameter updates or domain adaptation.

TABLE IV

COMPARISON OF REGIONAL OBJECT CLASSIFICATION RESULTS FOR VARIOUS METHODS ON THE DIOR-RSVG DATASET (ACCURACY IN %)

Type(model)	Model	SS	SIoU
G	Qwen	74.5	60.5
	MGPT	8.3	6.0
	LLaVa	72.6	55.9
G&VP	Sphinx	93.7	89.4
G&TF	MILS	73.2	60.4
G&VP&TF	ConMLLM	76.1	65
RS	GeoC	77.5	67
	LHRS	79.6	68.8
RS&VP	EMker	**98.37**	**97.24**
RS&VP&TF	TMLLM	92.4	88.6

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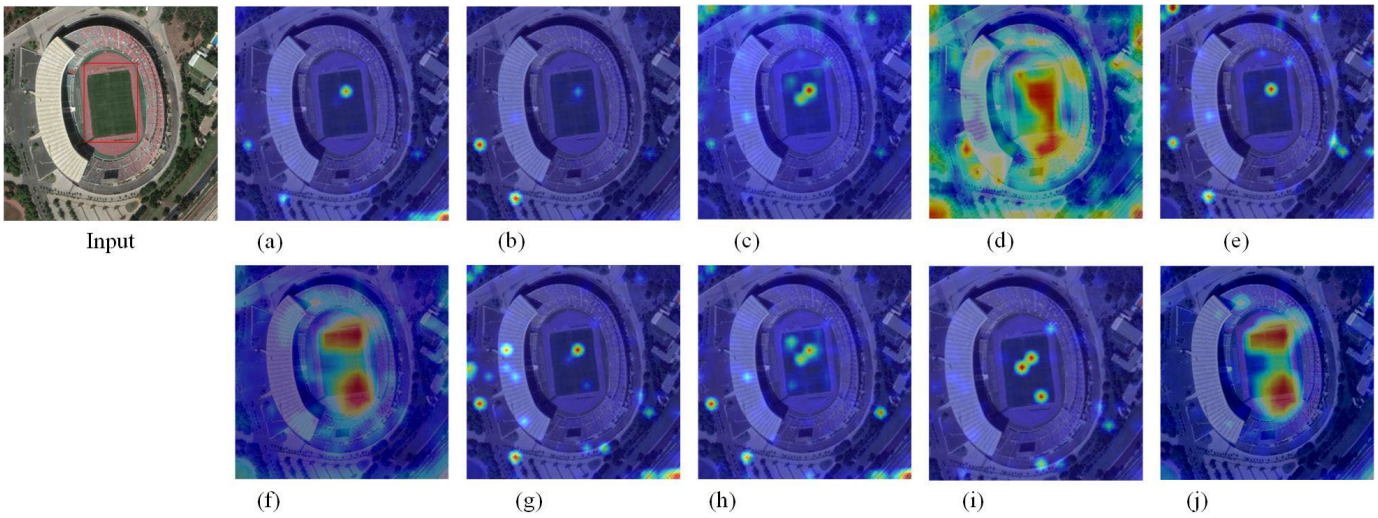


Fig. 3. Visualized attention map of stadium using different approaches on DIOR-RSVG dataset. (a) QwenVL-Chat. (b) MiniGPTV2. (c) LLaVA-1.5. (d) Sphinx. (e) MILS. (f) ControlMLLM. (g) GeoChat. (h) LHRS-Bot-Nova. (i) EarthMarker. (j) TFVP-MLLM.

To examine the SS and SIoU metrics more closely, the distinction between methods becomes most apparent in the SIoU score. EarthMarker achieves 97.24% compared to 88.6% for our approach, a difference that reflects their underlying mechanisms. The supervised models typically employ regression heads or segmentation decoders optimized by loss functions to generate hard and pixel-sharp boundaries. In contrast, our TFVP-MLLM derives regions from latent attention maps, which serve as soft probability distributions for semantic relevance. While these maps accurately encompass the target object, their boundaries are naturally smoother and less geometrically rigid than those produced by supervised regression, leading to a lower pixelwise overlap score despite correct localization. Regarding the SS metric, TFVP-MLLM achieves a competitive score of 92.4%. This result is comparable to other advanced methods like Sphinx and significantly outperforms general MLLMs, such as Qwen-VL-Chat and LLaVA-1.5. The relatively minor gap between our method and EarthMarker in this metric demonstrates that our model successfully captures the semantic essence and centroid of the target objects. In summary, our TFVP-MLLM achieves competitive performance comparable to these supervised approaches with zero training cost in regional object classification.

2) Qualitative Analysis: To comprehensively demonstrate the scene classification performance of TFVP-MLLM, we conduct a qualitative comparative analysis with other methods and present the visualization results using samples from the DIOR-RSVG dataset in Fig. 3.

(a) Nonvisual prompting general-purpose MLLMs (G): Qwen-VL-Chat, MiniGPTV2, LLaVA-1.5, and MILS exhibit notably constrained attention distributions, characterized by spot-like patterns concentrated in limited central area of the stadium. This suggests that they struggle to comprehend the complex spatial details of remote sensing imagery.

b) Visual prompting general-purpose MLLMS (G&VP): ControlMLLM yields the most prominent heatmap response with wide coverage and high intensity demonstrating its comprehensive understanding of the stadium's overall structure. Meanwhile, with distinct regional hot spots focused on the central field and stadium stands, Sphinx demonstrates more concentrated and structured attention distribution.

c) Nonvisual prompting remote sensing-specialized MLLMS (RS): GeoChat presents a scattered attention distribution, with its focus predominantly concentrated along the stadium perimeter. This pattern reflects strong boundary perception but indicates limited comprehension of the stadium's internal structures. The linearly dispersed hotspot pattern in LHRS-Bot-Nova implies a concurrent ability for functional zone discrimination and a deficiency in focused attention. Despite their domain expertise, the absence of visual prompts severely limits the attentional efficiency of both models.

d) Visual prompting remote sensing-specialized MLLMS (RS&VP): The heatmap of EarthMarker reveals a stronger attention to the central area of the stadium. In contrast, TFVP-MLLM accurately identifies the stadium's oval structure with clear boundaries and low noise. This demonstrates that our TFVP-MLLM exhibits superior performance by enabling the LLM's attention distribution to closely align with the target object.

3) Visualization Analysis of Attention Map: To demonstrate how text prompts and HGRAL optimize the LLM's attention distribution, we visualize the effects of different processing approaches across multiple layers. As depicted in Fig. 4, the model's attention distribution over the image can be clearly influenced by different text prompts. When no specific target is given in the text prompt, attention is uniformly spread across the entire image along with a general description. When a target is specified, the concentration of the attention map on the relevant region indicates the model's ability to capture semantic information from the image. As shown in Fig. 4 (3), incorporating HGRAL to optimize the latent visual variables concentrates the attention around the "airplane." This effective

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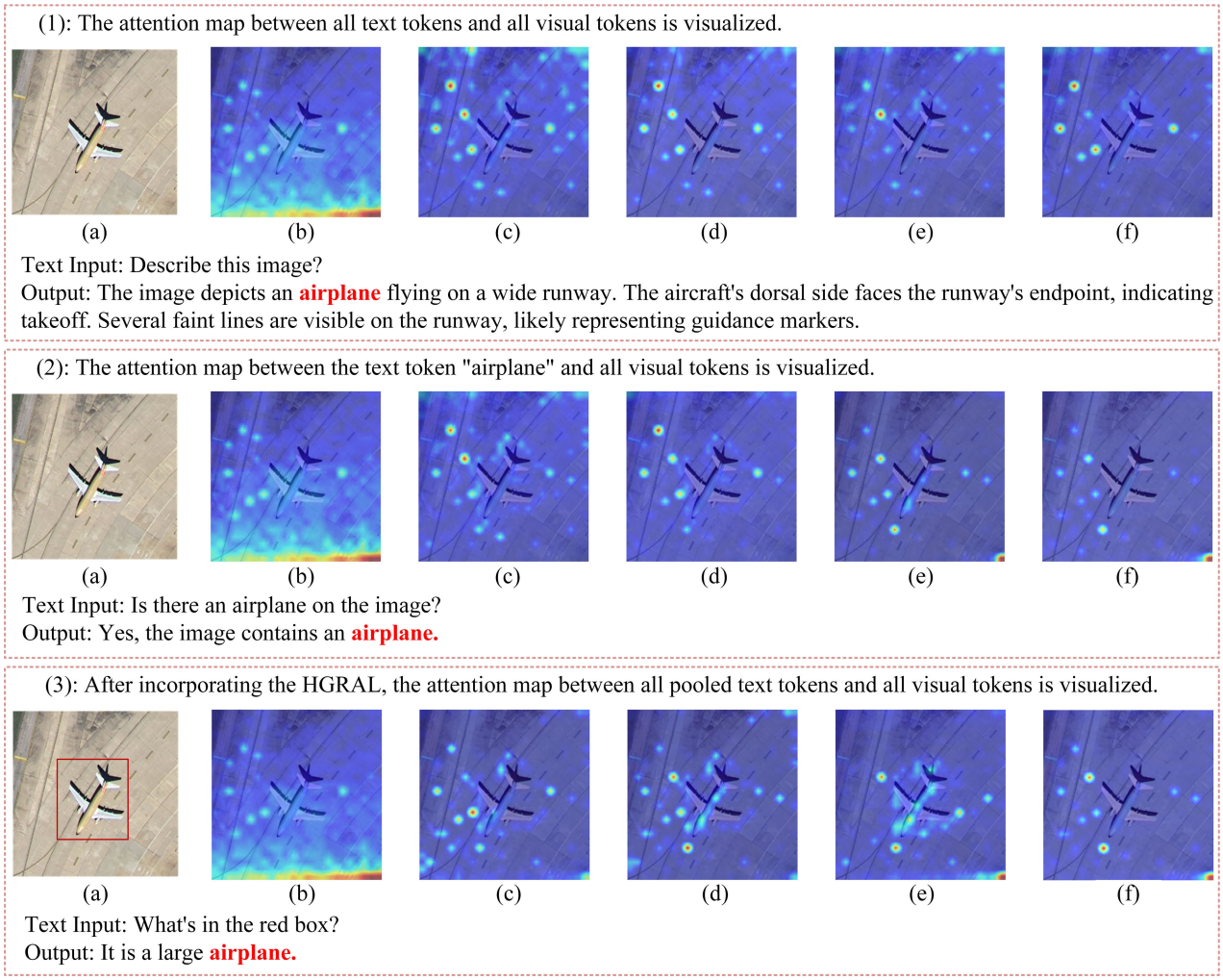


Fig. 4. Visualization of attention distributions across layers with different text prompts. (a) Input image. (b) Attention map from the 0th head. (c) Attention map from the 5th head. (d) Attention map from the 15th head. (e) Attention map from the 25th head. (f) Attention map from the 31th head.

guidance confirms that both text prompts and HGRAL can deliberately direct the model's attention.

4) Computational Complexity Analysis: To illustrate the computation complexities of all comparative methods, we report inference time, GPU memory usage (MiB), and classification accuracy in Figs. 5 and 6. The results demonstrate that TFVP-MLLM achieves highly competitive accuracy across all datasets while maintaining significantly lower GPU memory usage compared to other methods. This efficiency stems from the strategy of updating only the latent visual variables during inference, which incurs minimal overhead compared to the baseline. In particular, TFVP-MLLM offers inference speeds on par with EarthMarker, with substantially lower GPU memory usage, better scene classification performance, and only a slight cost to regional object accuracy. Furthermore, TFVP-MLLM significantly outperforms Sphinx in scene classification accuracy at a consistently lower memory usage. Despite requiring longer inference times due to its multiple rounds of generation-scoring feedback, TFVP-MLLM still achieves excellent performance across all datasets while maintaining low GPU memory usage. Consequently, our TFVP-MLLM is more appropriate in practical applications with respect to the balance of accuracy and computation cost. Additionally, a variant is introduced by incorporating the artificially encoded attention factor η into the visual token attention maps of reference prompt regions.

D. Ablation Experiments

1) HGRAL: The goal of HGRAL is to direct attention toward visually prompted regions by optimizing latent visual variables in attention maps under training-free schema. To evaluate its effectiveness, we conduct ablation experiments using variants featuring different granularity losses, alongside additional explorations on edit attention factors [$\eta = 5, 10, 20$ from (9)].

As seen in Table V and Fig. 7, the quantitative analysis demonstrates that the fusion of $\mathcal{L}_{\text{image}}$, $\mathcal{L}_{\text{obj}}$, and $\mathcal{L}_{\text{pixel}}$ yields a 7.9% improvement (84.5%→ 92.4%) over SS compared to the baseline that only uses $\mathcal{L}_{\text{image}}$. Notably, the two variants employing HGRAL with edit attention factors ($\eta = 5$ and $\eta = 20$) achieve inferior results with accuracies of only 60.9% and 18.7%, respectively. As shown in Fig. 7, a low η provides insufficient region control for correct classification, while a high value causes a collapse in linguistic capability

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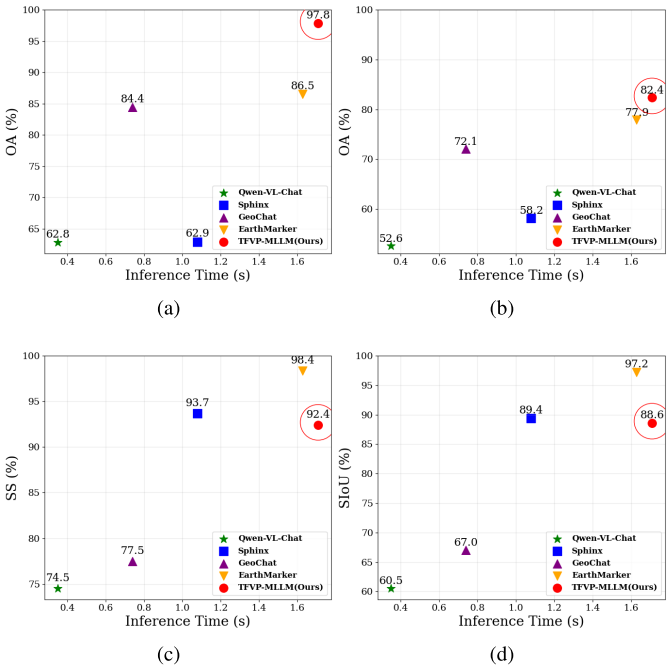


Fig. 5. Classification accuracy and inference time of each comparative method on the three datasets. Our proposed TFVP-MLLM is highlighted by red circle. (a) OA (%) from UCM dataset. (b) OA (%) from AID dataset. (c) SS (%) from DIOR-RSVG dataset. (d) SIoU (%) from DIOR-RSVG dataset.

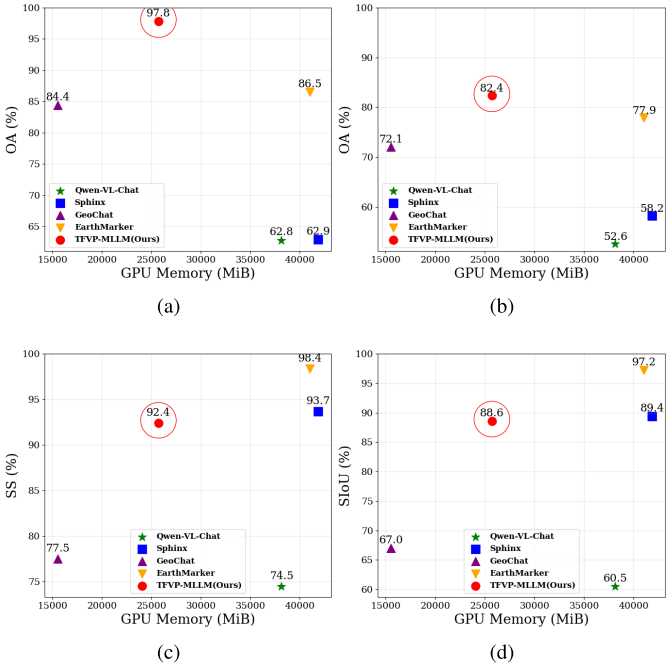


Fig. 6. Classification accuracy and CPU memory of each comparative method on the three datasets. Our proposed TFVP-MLLM is highlighted by red circle. (a) OA (%) from UCM dataset. (b) OA (%) from AID dataset. (c) SS (%) from DIOR-RSVG dataset. (d) SIoU (%) from DIOR-RSVG dataset.

that produces nonsensical outputs like "17." In contrast, the proposed HGRAL with $\eta = 10$ demonstrates superiority by effectively focusing on the prompted regions with a well-designed attention scale.

TABLE V

ABLATION STUDY ON HGRAL OF THE DIOR-RSVG DATASET (ACCURACY IN %)

$\mathcal{L}_{\text{image}}$	$\mathcal{L}_{\text{obj}}$	$\mathcal{L}_{\text{pixel}}$	$\eta = 5$	$\eta = 20$	SS
✓	✗	✗	✗	✗	84.5
✗	✓	✗	✗	✗	90.6
✗	✗	✓	✗	✗	86.2
✓	✓	✓	✗	✗	**92.4**
✗	✗	✗	✓	✗	60.9
✗	✗	✗	✗	✓	18.7

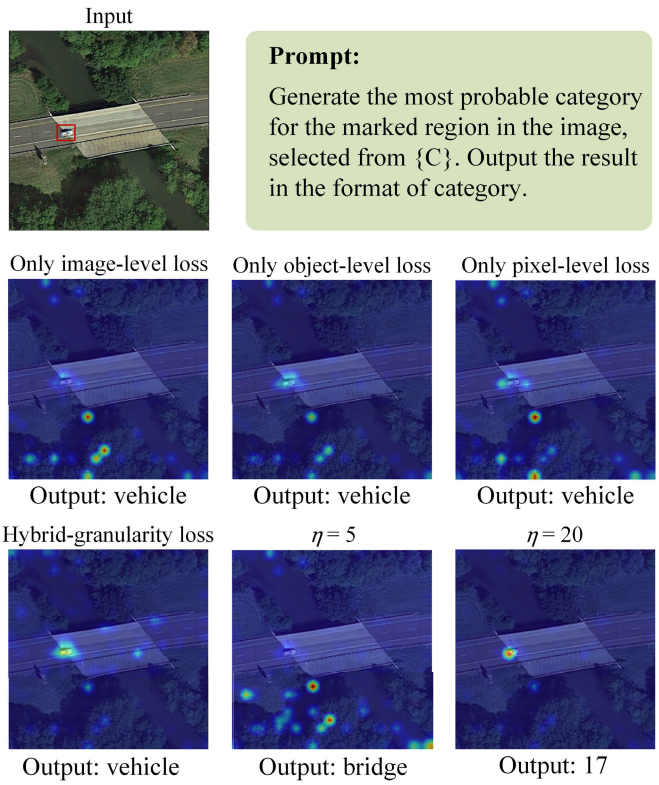


Fig. 7. Qualitative results of ablation study on HGRAL.

TABLE VI

ABLATION STUDY ON THE VISUAL SCORER IN SFR. UCM AND AID REPORT OA AND DIOR-RSVG REPORTS SS (ACCURACY IN %)

Scorer	UCM	AID	DIOR-RSVG
CLIP	97.8	82.4	92.4
RemoteCLIP	97.9	84.7	93.5

2) Visual Scorer in SFR: To investigate the influence of domain-specific visual representations on the scoring stage, we replace the general-purpose CLIP scorer with the remote sensing-specific RemoteCLIP [21] within SFR, where both models adopt the ViT-L/14 backbone. The comparative results are presented in Table VI. Two key observations can be drawn from this comparison. The general-purpose CLIP scorer already achieves competitive performance, while Remote-

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TABLE VII

ABLATION STUDY ON EACH MODULE OF TFVP-MLLM ON THE UCM, AID, AND DIOR-RSVG DATASETS. UCM AND AID REPORT OA AND DIOR-RSVG REPORTS SS (ACCURACY IN %)

GPIM	HGRAL	SFR	UCM	AID	DIOR-RSVG
✗	✗	✗	71.8	62.7	79.6
✓	✗	✗	86.1	73.8	86.2
✗	✓	✗	77.4	65.9	90.3
✗	✗	✓	84.2	72.3	83.9
✓	✓	✓	**97.8**	**82.4**	**92.4**

CLIP provides only marginal improvements (+0.1% on UCM, +2.3% on AID, and +1.1% on DIOR-RSVG). This suggests that GPIM and HGRAL have effectively narrowed the domain gap prior to the scoring stage, reducing the scorer's role from open-domain recognition to relative ranking among a small set of plausible candidates. Furthermore, the framework exhibits scorer-agnostic flexibility, as RemoteCLIP can serve as a plug-and-play replacement without any architectural modification. The most notable improvement appears on the AID dataset, where the multiresolution variations benefit from remote sensing-specific visual features.

3) Each Module of the Proposed TFVP-MLLM: To investigate the importance and effectiveness of each module, we selectively add each module to conduct ablation experiments and present the ablation results in Table VII. The baseline attains an accuracy of 79.6% when GPIM, HGRAL, and SFR are removed from the default TFVP-MLLM. On DIOR-RSVG, HGRAL demonstrates the most substantial SS improvement (+10.7% and 79.6%→ 90.3%) by adaptively regulating attention weights in visual-prompted regions, confirming that precise spatial attention is critical for regional object classification where fine-grained object localization is required. Across all three datasets, GPIM and SFR contribute improvements of comparable magnitude, with GPIM consistently yielding slightly larger gains (e.g., +14.3% versus +12.4% on UCM, +11.1% versus +9.6% on AID, and +6.6% versus +4.3% on DIOR-RSVG). This indicates that geographic prior injection and iterative scoring feedback provide complementary yet similarly effective benefits for both scene and regional object classification. Regarding HGRAL, its behavior differs between the two tasks. In scene classification, the mask covers the entire image, which causes $\mathcal{L}_{\text{image}}$ to become zero and $\mathcal{L}_{\text{pixel}}$ to reduce to a spatial constant, as analyzed in Section III-B. Consequently, only $\mathcal{L}_{\text{obj}}$ remains active and encourages a uniform attention distribution that promotes holistic scene-level feature extraction rather than local fixation. Although this represents a weaker optimization signal compared to the full three-component HGRAL employed in regional object classification, it still produces measurable gains of +5.6% on UCM and +3.2% on AID. In regional object classification, where all three loss components are active, HGRAL delivers the largest single-module improvement (+10.7% on DIOR-RSVG). Therefore, the above observations and discussions underpin the necessity of these three components when constructing our proposed TFVP-MLLM.

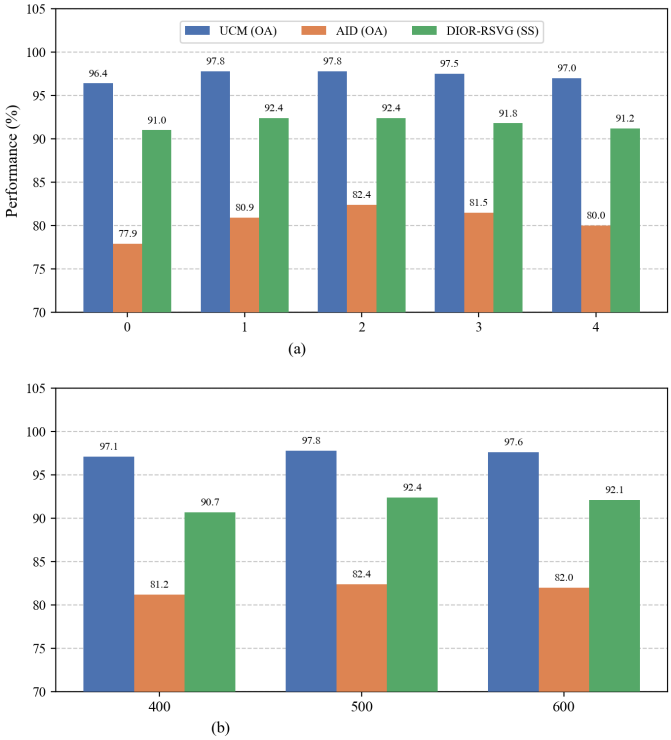


Fig. 8. Ablation study on (a) threshold θ_{sim} and (b) guidance strength control parameter α.

4) Ablation Study on Critical Parameters: In this section, we first investigate the threshold θ_{sim} in (15) and the guidance strength control parameter α. As illustrated in Fig. 8(a), θ_{sim} determines whether a generated candidate R^i is corrected to the best-matching category C^{best} based on the maximum SS score. As observed, the proposed method achieves optimal performance on the AID and DIOR-RSVG datasets when $\theta_{sim} = 0.5$, yielding OAs of 82.4% and 92.4%, respectively, while the UCM dataset peaks at 97.8% within the range of 0.4 to 0.5. A lower threshold (e.g., < 0.4) may introduce erroneous semantic corrections, whereas a higher threshold (e.g., > 0.5) imposes overly strict filtering that limits effective candidate rectification. Consequently, we set $\theta_{sim} = 0.5$ as the optimal configuration for subsequent experiments.

Besides, we investigate the influence of the guidance strength control parameter α ranging from 400 to 600 on the OA for the UCM and AID datasets and on the SS for the DIOR-RSVG dataset. Since α acts as a hyperparameter governing the update of learnable latent variables within the attention map, its variation exhibits a trend analogous to that of the hyperparameters J_1 and J_2. Specifically, an increase in α amplifies the update step size of the attention map, thereby affecting the model's interpretation of key regions. Experimental results demonstrate that $\alpha = 500$ yields the optimal performance across all three datasets. An insufficient α leads to inadequate attention map updates and fails to capture critical features, whereas an excessive value causes overly rapid updates that destabilize model convergence. Consequently, $\alpha = 500$ is selected as the optimal configuration.

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V. Conclusion

To address the challenges of inaccurate region localization, limited domain adaptability, and high computational costs associated with the existing MLLM-based methods, we propose a training-free MLLM approach named TFVP-MLLM with visual prompting guidance. Particularly, the CPG is responsible for generating high-quality candidate categories, and the SFR uses the iterative generation-scoring feedback to gradually correct the classification results and continuously improve the classification accuracy. As two core components of CPG, GPIM enhances scene understanding by integrating global geographic knowledge into text prompts, while HGRAL improves key region perception by employing a hybrid-granularity loss to optimize visual tokens within MLLM attention maps. Consequently, the proposed TFVP-MLLM achieves the state-of-the-art performance among training-free multimodal approaches in remote sensing image classification on the UCM, AID, and DIOR-RSVG datasets in an efficient way.

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Yifei Xu received the B.S. degree in computer science and technology, from South China University of Technology (SCUT), Guangzhou, China, in 2011, and the Ph.D. degree from the School of Computer Science, Zhejiang University, HangZhou, China, in 2017.

Since 2021, he has been with the School of Software of Xi'an Jiaotong University, Xi'an, China, as an Associate Professor. His current research interests include hyperspectral image processing and video understanding.



Zixiang Ni received the B.E. degree in data science and big data technology from China Agricultural University, Beijing, China, in 2025. He is currently pursuing the M.S. degree with the School of Software Engineering, Xi'an Jiaotong University, Xi'an, China.

His research interests lie primarily in computer vision and video understanding.



Pingping Wei received the B.S. degree in computer science and technology, Tianjin University, Tianjin, China, in 2012, and the M.S. degree from The University of Sydney, Sydney, Australia, in 2016.

She is currently working with Xi'an Jiaotong University, Xi'an, China.

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Chufan Xiao currently pursuing a joint double Bachelor of Architecture degree from Xi'an Jiaotong University, Xi'an, China, and Politecnico di Milano, Milan, Italy, along with a Bachelor of Science degree in Computer Science from Xi'an Jiaotong University.

His main areas of interest lie in computer vision and full-stack development.



Yaochen Li received the Ph.D. degree in pattern recognition and intelligent system from Xi'an Jiaotong University, Xi'an, China, in 2016.

From December 2012 to January 2014, he was a Visiting Scholar with IPAM, University of California at Los Angeles, Los Angeles, CA, USA. He is currently an Associate Professor with the School of Software, Xi'an Jiaotong University. His research interests include image processing and analysis, computer vision, and intelligent transportation systems.



Agen Qiu joined Chinese Academy of Surveying and Mapping (CASM) in 2004, where he currently serves as a Research Fellow. He has long been dedicated to research in the fields of geographic information services, spatiotemporal big data governance, and spatiotemporal intelligence.



Dong Li received the B.E. and M.S. degrees from the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an, China, in 2013 and 2018, respectively.

He has been an Engineer with Xi'an Universal Satellite TT&C and Data Application Company Ltd., Xi'an, China. His research interests lie primarily in artificial intelligence technology in the field of satellite data processing.



Yuan Rao is a Professor with the School of Software, Xi'an Jiaotong University, Xi'an, China, where he is the Director of Xi'an Key Laboratory for Social Intelligence and Complex Data Processing. His research interests include social intelligence and sentiment analysis, algorithmic trading and assets pricing, knowledge graph and service, and complexity data processing.

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